

```
Activities Terminal May 21 23:16 vy23521830@UIT: -
vy23521830@UIT:~$ pyspark
[I 2026-05-21 23:16:54.163 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-05-21 23:16:54.167 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-05-21 23:16:54.175 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-05-21 23:16:54.184 ServerApp] notebook | extension was successfully linked.
[I 2026-05-21 23:16:54.989 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-05-21 23:16:55.071 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-05-21 23:16:55.074 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-05-21 23:16:55.076 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-05-21 23:16:55.092 LabApp] JupyterLab extension loaded from /home/vy23521830/.local/lib/python3.8/site-packages/jupyterlab
[I 2026-05-21 23:16:55.092 LabApp] JupyterLab application directory is /home/vy23521830/.local/share/jupyter/lab
[I 2026-05-21 23:16:55.094 LabApp] Extension Manager is 'pypl'.
[I 2026-05-21 23:16:55.228 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-05-21 23:16:55.234 ServerApp] notebook | extension was successfully loaded.
[I 2026-05-21 23:16:55.235 ServerApp] Serving notebooks from local directory: /home/vy23521830
[I 2026-05-21 23:16:55.235 ServerApp] Jupyter Server 2.14.2 is running at:
[I 2026-05-21 23:16:55.235 ServerApp] http://localhost:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
[I 2026-05-21 23:16:55.235 ServerApp] http://127.0.0.1:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
[I 2026-05-21 23:16:55.235 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-05-21 23:16:55.364 ServerApp]

To access the server, open this file in a browser:
file:///home/vy23521830/.local/share/jupyter/runtime/jpserver-2083-open.html
Or copy and paste one of these URLs:
http://localhost:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
http://127.0.0.1:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
[I 2026-05-21 23:16:55.440 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server,
r-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, p
ython-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-lan
guageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yaml-language-server
```

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[I 2026-05-21 23:16:54.175 ServerApp] jupyterlab | extension was successfully linked.
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[I 2026-05-21 23:16:55.092 LabApp] JupyterLab application directory is /home/vy23521830/.local/share/jupyter/lab
[I 2026-05-21 23:16:55.094 LabApp] Extension Manager is 'pypl'.
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[I 2026-05-21 23:16:55.235 ServerApp] http://127.0.0.1:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
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file:///home/vy23521830/.local/share/jupyter/runtime/jpserver-2083-open.html
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http://localhost:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
http://127.0.0.1:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
[I 2026-05-21 23:16:55.440 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server,
r-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, p
ython-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-lan
guageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yaml-language-server
[W 2026-05-21 23:17:16.107 ServerApp] 404 GET /lab/api/settings/@jupyterlab/apputils-extension:kernels-settings?1779380236089 (127.0
.0.1): Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/apputils-extension:kernels-settings.json
[W 2026-05-21 23:17:16.107 LabApp] wrote error: 'Schema not found: /home/vy23521830/.local/share/jupyter/lab/schemas/@jupyterlab/app
utils-extension:kernels-settings.json'
Traceback (most recent call last):
  File "/home/vy23521830/.local/lib/python3.8/site-packages/tornado/web.py", line 1788, in _execute
    result = method(*self.path_args, **self.path_kwargs)
  File "/home/vy23521830/.local/lib/python3.8/site-packages/tornado/web.py", line 3301, in wrapper
    return method(self, *args, **kwargs)
```

```
Activities Terminal May 21 23:25 vy23521830@UIT: ~  
vy23521830@UIT: ~  
26/05/21 23:23:41 WARN JavaUtils: Attempt to delete using native Unix OS command failed for path = /tmp/spark-283cddb-b37d-4c89-8e7  
7-e50b61067722/userFiles-8a117f57-d7c4-46fa-8fa6-422b00a28cf2. Falling back to Java IO way  
java.io.IOException: Failed to delete: /tmp/spark-283cddb-b37d-4c89-8e77-e50b61067722/userFiles-8a117f57-d7c4-46fa-8fa6-422b00a28cf  
2  
at org.apache.spark.network.util.JavaUtils.deleteRecursivelyUsingUnixNative(JavaUtils.java:174)  
at org.apache.spark.network.util.JavaUtils.deleteRecursively(JavaUtils.java:109)  
at org.apache.spark.network.util.JavaUtils.deleteRecursively(JavaUtils.java:90)  
at org.apache.spark.util.SparkFileUtils.deleteRecursively(SparkFileUtils.scala:121)  
at org.apache.spark.util.SparkFileUtils.deleteRecursively$(SparkFileUtils.scala:120)  
at org.apache.spark.util.Utils$.deleteRecursively(Utils.scala:1126)  
at org.apache.spark.util.ShutdownHookManager$.anonfun$new$4(ShutdownHookManager.scala:65)  
at org.apache.spark.util.ShutdownHookManager$.anonfun$new$4$adapted(ShutdownHookManager.scala:62)  
at scala.collection.IndexedSeqOptimized.foreach(IndexedSeqOptimized.scala:36)  
at scala.collection.IndexedSeqOptimized.foreach$(IndexedSeqOptimized.scala:33)  
at scala.collection.mutable.ArrayOps$ofRef.foreach(ArrayOps.scala:198)  
at org.apache.spark.util.ShutdownHookManager$.anonfun$new$2(ShutdownHookManager.scala:62)  
at org.apache.spark.util.ShutdownHookManager$.run(ShutdownHookManager.scala:214)  
at org.apache.spark.util.ShutdownHookManager$.anonfun$runAll$2(ShutdownHookManager.scala:188)  
at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)  
at org.apache.spark.util.Utils$.logUncaughtExceptions(Utils.scala:1928)  
at org.apache.spark.util.ShutdownHookManager$.anonfun$runAll$1(ShutdownHookManager.scala:188)  
at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)  
at scala.util.Try$.apply(Try.scala:213)  
at org.apache.spark.util.ShutdownHookManager$.runAll(ShutdownHookManager.scala:188)  
at org.apache.spark.util.ShutdownHookManager$.anonfun$run$2(ShutdownHookManager.scala:178)  
at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:515)  
at java.base/java.util.concurrent.FutureTask.run(FutureTask.java:264)  
at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1128)  
at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:628)  
at java.base/java.lang.Thread.run(Thread.java:829)  
[I 2026-05-21 23:23:47.010 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-03/23521830_Lab03.ipynb  
26/05/21 23:23:48 WARN Utils: Your hostname, UIT resolves to a loopback address: 127.0.1.1; using 10.0.2.15 instead (on interface en  
p0s3)  
26/05/21 23:23:48 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
```

```
Activities Terminal May 21 23:24 vy23521830@UIT: ~  
vy23521830@UIT: ~  
[I 2026-05-21 23:18:16.933 ServerApp] Connecting to kernel 7f6f302d-8baf-4a18-b546-df05ce6b2010.  
[I 2026-05-21 23:18:16.997 ServerApp] Connecting to kernel 7f6f302d-8baf-4a18-b546-df05ce6b2010.  
[I 2026-05-21 23:19:46.867 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-03/23521830_Lab03.ipynb  
[I 2026-05-21 23:23:02.777 ServerApp] Kernel restarted: 7f6f302d-8baf-4a18-b546-df05ce6b2010  
[I 2026-05-21 23:23:03.142 ServerApp] Starting buffering for 7f6f302d-8baf-4a18-b546-df05ce6b2010:15ba7cea-8d2d-438f-bcce-de0a55cfff  
ae  
[I 2026-05-21 23:23:03.302 ServerApp] Connecting to kernel 7f6f302d-8baf-4a18-b546-df05ce6b2010.  
[I 2026-05-21 23:23:03.306 ServerApp] Restoring connection for 7f6f302d-8baf-4a18-b546-df05ce6b2010:15ba7cea-8d2d-438f-bcce-de0a55cf  
ffae  
26/05/21 23:23:14 WARN Utils: Your hostname, UIT resolves to a loopback address: 127.0.1.1; using 10.0.2.15 instead (on interface en  
p0s3)  
26/05/21 23:23:14 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
26/05/21 23:23:20 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where  
applicable  
[I 2026-05-21 23:23:22.091 ServerApp] Kernel started: 315d00e4-791a-43ef-bab5-ae9b1bed2f01  
[I 2026-05-21 23:23:22.104 ServerApp] Kernel shutdown: 7f6f302d-8baf-4a18-b546-df05ce6b2010  
26/05/21 23:23:28 WARN Utils: Your hostname, UIT resolves to a loopback address: 127.0.1.1; using 10.0.2.15 instead (on interface en  
p0s3)  
26/05/21 23:23:28 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address  
Setting default log level to "WARN".  
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).  
26/05/21 23:23:32 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where  
applicable  
[I 2026-05-21 23:23:38.232 ServerApp] Connecting to kernel 315d00e4-791a-43ef-bab5-ae9b1bed2f01.  
[I 2026-05-21 23:23:41.245 ServerApp] Kernel started: abc1ce58-bd79-4a9f-899a-8540cdd71a5d  
[I 2026-05-21 23:23:41.248 ServerApp] Kernel shutdown: 315d00e4-791a-43ef-bab5-ae9b1bed2f01  
26/05/21 23:23:41 WARN JavaUtils: Attempt to delete using native Unix OS command failed for path = /tmp/spark-283cddb-b37d-4c89-8e7  
7-e50b61067722/userFiles-8a117f57-d7c4-46fa-8fa6-422b00a28cf2. Falling back to Java IO way  
java.io.IOException: Failed to delete: /tmp/spark-283cddb-b37d-4c89-8e77-e50b61067722/userFiles-8a117f57-d7c4-46fa-8fa6-422b00a28cf  
2  
at org.apache.spark.network.util.JavaUtils.deleteRecursivelyUsingUnixNative(JavaUtils.java:174)  
at org.apache.spark.network.util.JavaUtils.deleteRecursively(JavaUtils.java:109)  
at org.apache.spark.network.util.JavaUtils.deleteRecursively(JavaUtils.java:90)  
at org.apache.spark.util.SparkFileUtils.deleteRecursively(SparkFileUtils.scala:121)
```

```
Activities Terminal May 21 23:25 vy23521830@UIT: ~
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at scala.collection.IndexedSeqOptimized.foreach$(IndexedSeqOptimized.scala:33)
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at org.apache.spark.util.ShutdownHookManager$.anonfun$runAll$2(ShutdownHookManager.scala:62)
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at scala.runtime.java8.JFunction0$mcV$sp.apply(JFunction0$mcV$sp.java:23)
at org.apache.spark.util.Utils$.logUncaughtExceptions(Utils.scala:1928)
at org.apache.spark.util.ShutdownHookManager$.anonfun$runAll$1(ShutdownHookManager.scala:188)
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at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:628)
at java.base/java.lang.Thread.run(Thread.java:829)
[I 2026-05-21 23:23:47.010 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-03/23521830_Lab03.ipynb
26/05/21 23:23:48 WARN Utils: Your hostname, UIT resolves to a loopback address: 127.0.1.1; using 10.0.2.15 instead (on interface en
p0s3)
26/05/21 23:23:48 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/05/21 23:23:53 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where
applicable
[I 2026-05-21 23:23:56.300 ServerApp] Connecting to kernel abc1ce58-bd79-4a9f-899a-8540cdd71a5d.
[I 2026-05-21 23:25:14.281 ServerApp] Saving file at /BTTH_Spark/DS200/Lab-03/23521830_Lab03.ipynb
^C[I 2026-05-21 23:25:27.441 ServerApp] interrupted
[I 2026-05-21 23:25:27.442 ServerApp] Serving notebooks from local directory: /home/vy23521830
1 active kernel
Jupyter Server 2.14.2 is running at:
http://localhost:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
http://127.0.0.1:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
Shut down this Jupyter server (y/[n])? y
[C 2026-05-21 23:25:29.995 ServerApp] Shutdown confirmed
[I 2026-05-21 23:25:29.996 ServerApp] Shutting down 5 extensions
```

```
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http://127.0.0.1:8888/tree?token=b6700889d85157c231c3414689c43ac7f51a7ea17194b72e
Shut down this Jupyter server (y/[n])? y
[C 2026-05-21 23:25:29.995 ServerApp] Shutdown confirmed
[I 2026-05-21 23:25:29.996 ServerApp] Shutting down 5 extensions
[I 2026-05-21 23:25:29.998 ServerApp] Shutting down 1 kernel
[I 2026-05-21 23:25:30.004 ServerApp] Kernel shutdown: abc1ce58-bd79-4a9f-899a-8540cdd71a5d
26/05/21 23:25:30 WARN JavaUtils: Attempt to delete using native Unix OS command failed for path = /tmp/spark-6a6e18ac-5e09-489a-aea
c-8f96198c743f. Falling back to Java IO way
java.io.IOException: Failed to delete: /tmp/spark-6a6e18ac-5e09-489a-aeac-8f96198c743f
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at org.apache.spark.util.ShutdownHookManager$.anonfun$runAll$1(ShutdownHookManager.scala:188)
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at java.base/java.util.concurrent.Executors$RunnableAdapter.call(Executors.java:515)
at java.base/java.util.concurrent.FutureTask.run(FutureTask.java:264)
at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1128)
at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:628)
at java.base/java.lang.Thread.run(Thread.java:829)
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